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# The impact of new metro on travel behavior: Panel analysis using mobile phone data

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ABSTRACT

The impact of urban rail transit on changes in various travel behavior outcomes has long been debated. However, few studies have used large-scale panel data to investigate how urban rail transit affects travel behavior. In the case of the new metros in Shenzhen, China, we constructed a four-wave panel of 7,799 residents across two years by using mobile phone data. We used a treatment–control group research design and a difference-in-differences analysis to evaluate the impact of the new metros on travel behavior in catchment areas of 0–1, 1–2, and 2–3 km, respectively. The results show that the appearance of the new metros increased the metro trips and activity space for the residents living in the 0–3 km range, but only increased total trip frequency and distance for the residents living in the 0–1 km range. New metro trips mainly replaced bike, e-bike, and bus trips, and slowed the growth trend of car trips. The findings help planners and policymakers better understand the impact of urban rail transit on the residents' mobilities.

## 1. Introduction

Many large cities in developing countries have experienced rapid urbanization in recent decades, resulting in dramatic urban spatial change. Urban spatial change, such as urban sprawl, is usually accompanied by changes in peoples' housing and employment locations, which leads to longer travel times and distances, as well as challenges such as air pollution, social justice, and transport equity (Crane, 2000). Urban rail transit provides people who rely on public transit a fast and reliable transportation mode while also promoting environmental sustainability and social equity (Shen et al., 2016). Investing in urban rail transit and transit-oriented development (TOD) near urban rail transit stations is regarded as a key solution for large cities to address these challenges (Cao and Cao, 2014). Despite rapid expansion, urban rail transit is a durable good with a large initial investment and long-term operating subsidies (Zhang, 2020), thus assessing the travel behavior benefits of urban rail transit systems on residents is important for urban rail transit investments (Zhao et al., 2018).

Many studies have investigated the impact of urban rail transit on travel behavior (Ewing and Cervero, 2010) and a large body of evidence shows that urban rail transit makes people living nearby to use urban rail transit more and drive cars less (Arrington and Cervero, 2008). However, some research gaps are yet to be explored. First, few studies have examined the impact of urban rail transit on people's total trips, i.e., does urban rail transit induce people to take more or longer trips, and on people's activity space, i.e., where

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and how their travels take place, and the source of increased metro trips, particularly whether metro trips can replace car trips. Second, most studies used 800 m or 1 km as the catchment area of urban rail transit stations (Guerra et al., 2012; Spears et al., 2017), which may be small, especially in the era of shared micro-mobility. If urban rail transit stations have larger catchment areas, how different are the travel behaviors of residents living in different distance catchment areas? Third, and most importantly, these studies rely heavily on cross-sectional survey data, which may pose challenges in making causal inferences because of the lack of a temporal dimension (Dai et al., 2020). Panel survey data are more suitable for making causal inferences, but their use in relevant studies is very limited. The use of panel survey data may suffer from the problem of sample bias because the sample sizes of panel survey data are usually very small (Spears et al., 2017). Mobile phone data, which are becoming more prevalent in travel behavior studies, have the panel nature that can track travel behavior changes over time among the same group of people (Wang et al., 2017). However, to the best of our knowledge, no study has used mobile phone data to construct genius panel data in travel behavior studies.

We aim to fill these research gaps with this study. Using the new metros in Shenzhen, China as a case study, we used multi-wave and large-scale genius panel data to answer how do residents living in the station areas change their travel behaviors. The contributions of this study lie in the following distinguishing features: 1) the use of mobile phone data to construct genius panel data which cover a large sample of people and record their long-term travel behaviors; 2) the use of longitudinal design and difference-in-differences analysis to assess the impact of new metros on travel behavior, particularly total trips and activity space, as well as the source of increased metro trips; 3) the use of large catchment areas and the unraveling of travel behaviors of residents living in different distance catchment areas.

The rest of the paper is structured as follows: Section 2 provides a literature background; Section 3 presents the data; Section 4 introduces the methodology; Section 5 analyzes the results; Section 6 discusses the empirical findings; the last section makes conclusions.

## 2. Literature review

The impact of urban rail transit and its surrounding built environment on travel behavior has been studied extensively. This section focuses solely on the impact of new urban rail transit on travel behavior and reviews the empirical findings and the data used in relevant research, as well as the use of mobile phone data in travel behavior studies.

### *2.1. Empirical findings of urban rail transit impact studies*

Previous studies have examined the relationship between urban rail transit and travel behavior outcomes, particularly urban rail transit use and car use. The result that new urban rail transit services are associated with more urban rail transit trips has long been recognized (Cervero, 1994; Lund et al., 2006; Shen et al., 2016; Brown and Werner, 2007, 2008; Spears et al., 2017). The impact of new urban rail transit on urban rail transit trips is usually greater for commuting trips than non-commuting trips (Lund et al., 2006), and it also varies considerably between cities (Cervero, 1994). Some studies have revealed that urban rail transit reduces car use, as measured by car trips (Brown and Werner, 2007, 2008; Pan et al., 2013), vehicle kilometers traveled (Houston et al., 2015; Spears et al., 2017; Cao, 2019), and fuel consumption (Zhang et al., 2017). The reduction in vehicle kilometers traveled by urban rail transit is significant, reaching 20% (Cao, 2019) or 10 miles per day (Spears et al., 2017). The impact of new urban rail transit on total trips and places visited (also known as activity spaces) has been less explored. Dai et al. (2020) found an insignificant impact of new urban rail transit on total trip frequency. Botte and Olaru (2012) observed an increase in activity spaces measured as kernel density with the appearance of a new urban rail transit line. However, Tao et al. (2020) found that despite the urban rail transit expansion and land use development, low-income people's total trip distance decreased and activity spaces measured as standard distance circles remained unchanged in Hong Kong, China between 2002 and 2011. Of course, this result cannot be attributed solely to the expansion of urban rail transit.

When investigating the impact of urban rail transit, the selection of treatment and control areas has a large impact on the results. Earlier studies used residents of whole cities or counties as controls (Cervero, 1994; Lund et al., 2006), while recent studies used residents living in comparable urban corridors as controls to avoid exaggerating the impact of urban rail transit because urban rail transit is usually built in areas that already had superior transit services (Cao, 2019). In most studies, a buffer zone of 800 m or 1 km in a straight line or network distance (15-minute walk) is used as a treatment area (Guerra et al., 2012; Spears et al., 2017). Although some studies have suggested that urban rail transit may have a wider range of impacts, few studies have used a larger buffer zone or multiple buffer zones of different distances as treatment areas. In addition, the use of residents outside of the 800 m or 1 km buffer zone as controls in some studies may underestimate the impact of urban rail transit.

### *2.2. Data used in urban rail transit impact studies*

These empirical findings are largely based on traditional survey data, especially cross-sectional survey data, which are gathered at a single point in time. Although frequently used in travel behavior studies, cross-sectional survey data are criticized for being unable to provide direct evidence about how people will change their travel behavior in response to infrastructure changes due to the lack of a temporal dimension, which prevents them from identifying the possible causal processes and may lead to misinformed decisions (Boarnet, 2011). Longitudinal analysis with panel data is more rigorous for examining the causal impact of infrastructure changes because panel data record the travel behavior changes of the same people in response to infrastructure changes. However, panel data are difficult to obtain in transportation research (Song et al., 2019). Only three travel behavior studies have used genius panel data to evaluate the impact of urban rail transit on travel behavior (Brown and Werner, 2007, 2008; Spears et al., 2017). Other two public

health studies found that new urban rail transit riders increased their physical activity which was associated with accessing/egressing stations while lowing their body mass index (BMI) (Brown et al., 2015; MacDonald et al., 2010). Since obtaining genius panel data for transportation research is difficult, other studies used longitudinal study design of repeated cross-sectional data (also known as pseudo panel data). Two studies using repeated cross-sectional data show that the appearance of a new urban rail transit line in Singapore results in a decrease in car mode share and an increase in rail mode share, either for high-income households (Zhu and Diao, 2016) or for the full sample (Dai et al., 2020).

Although there are many benefits of using panel data for travel behavior analyses, studies using panel data in the analysis of the impact of urban rail transit on travel behavior are rather rare. The sample size of genius panel data is usually small (a few hundred or a few dozen) because tracking respondents for years is difficult and expensive. The results are easily biased due to the small sample sizes, the dropouts of samples during the longitudinal survey process, and most importantly, the remaining sample becomes unrepresentative as attrition is not uniform across the sample. Moreover, there is daily, weekly, monthly regularity and irregularity in people's travel behavior (Järv et al., 2014), while traditional survey data only provide a snapshot of people's travel behavior within a very short time window, usually one weekday.

### 2.3. Mobile phone data in travel behavior studies

Big data, especially mobile phone data, have emerged and been used in travel behavior research (Chen et al., 2016; Wang et al., 2017). There are two types of mobile phone data: call detail record data and cellular signaling data. Call detail record data are solely based on phone usage, whereas cellular signaling data include all network-based records, in which user location is recorded on a regular basis, regardless of whether users are using their mobile phone or not (Fekih et al., 2020). The medians of the time intervals between consecutive data points are usually a few hours for call detail record data and a few minutes for cellular signaling data. Mobile phone data have pros and cons when compared with traditional survey data. On the one side, mobile phone data are collected continuously and automatically, covering larger samples of the population and recording detailed travel information. On the other side, mobile phone data are anonymous and usually do not have various sociodemographic attributes. In travel behavior research, mobile phone data have been used to identify travel behavior such as trip and activity purpose (Diao et al., 2016; Xiao et al., 2016), to explore travel behavior such as spatial and temporal distribution (Xu et al., 2015; Demissie et al., 2015; Zhao et al., 2020), to predict travel behavior such as OD matrix (Iqbal et al., 2014; Alexander et al., 2015; Toole et al., 2015; Ge and Fukuda, 2016), and to uncover the influencing factors of travel behavior (Picornell et al., 2015; Yang et al., 2019). However, to the best of our knowledge, no study has used mobile phone data for panel analysis, despite the fact that mobile phone data can zoom into and continually track individual travel behavior. Using mobile phone data to construct panel data can overcome the difficulties of data acquisition in traditional panel surveys, as well as capture travel behavior variability over time.

## 3. Data

The study site is Shenzhen, China. Shenzhen is China's first special economic zone and one of the four first-tier cities in China. Shenzhen has a total area of 1,997 $km^2$ and a permanent resident population of 17.56 million in 2020. In comparison to Western cities, particularly those in the United States, Asian cities, including Shenzhen, are featured by high-density urban contexts. These cities are facing transportation problems due to high-density urban sprawl during rapid urbanization. Shenzhen has built a large metro network, which has a total length of 422.6 km and 283 stations in 2020, to serve the mobility demands of its growing population. This study was designed to take advantage of the opening of metro lines 6 and 8 on August 18, 2020 to explore the impact of the new metro on travel behavior.

### 3.1. Data

We used mobile phone data from a leading communication operator with a market coverage of over 25% in Shenzhen. In this study, we used four months of cellular signaling data (November 2019, March 2020, November 2020, and March 2021). These four months are nine and five months before and three and seven months after the opening of the two new metro lines, respectively. The communication operator had processed cellular signaling data to provide detailed records of people's stationary activities and trips. Stationary activities were defined as a person staying in the same location for at least 30 min and trips were defined as moves between two stationary activities. The metro trips were identified based on whether the signal during the trip is connected to the metro system's solely owned signal towers. In our cellular signaling data, timing advance and lateration/triangulation were employed to improve location accuracy using low-level measurements such as received signal strength and transmission time of multiple cell towers in the available range, which have higher spatial resolution than previous studies that used the position of the cell tower connected by the mobile phone to represented the mobile phone's location (Jiang et al., 2013).

We focus on individuals who have not changed their residential locations during the four months. Individuals living within 0–1 km, 1–2 km, and 2–3 km network distance of the new metro stations of metro lines 6 and 8 were chosen as the three treatment groups, and individuals living further away but still along the metro corridors were chosen as the control group. Note that all of these individuals are within a 3 km straight-line distance buffer of the new metro stations to ensure that individuals in the control and treatment groups are all along the same metro corridor. A metro catchment area is usually determined by a threshold of actual or desired walking distance. Previous studies have typically used a buffer zone of 800 m or 1 km as a metro catchment area. However, bikes are a very popular feeder mode in China and a large proportion of metro riders are bicyclists at the start or end of their metro trips. According to

Zuo et al. (2018), the 85th percentile walking distance at home ends is 0.95 km, while the 85th percentile bicycling distance is 2.56 km. Lee et al. (2016) found that the distance of bike access to public transportation can be expanded up to 3 km. Thus we included buffer zones of 1–2 km, and 2–3 km network distances as treatment areas to explore whether the new metros have a wider range of impact. Fig. 1 shows the residential locations of 7,799 individuals used in this study. They are labeled in four different colors according to the network distances from their residential locations to the nearest new metro stations.

### 3.2. Indicators

Nine travel behavior indicators were generated using cellular signaling records (Table 1). Most indicators are self-explanatory. The standard distance circle is the circular area with a radius of standard distance. The calculation of the standard distance and standard distance circle follows the expressions:

$$Standard\ \ distance = \sqrt{\frac{\sum_{i=0}^{n}(x_i - \overline{X})^2}{n} + \frac{\sum_{i=0}^{n}(y_i - \overline{Y})^2}{n}} \tag{1}$$

$$Standard\ \ distance\ \ circle = \pi^* Standard\ \ distance^2 \tag{2}$$

where $x_i$ and $y_i$ are the projected coordinates of a location $i$; $\overline{X}$ and $\overline{Y}$ are the coordinates of the mean center of all locations; $n$ is the total number of locations. The minimum convex polygon develops the smallest possible polygon to delineate the spatial extensiveness of a person's activity locations. The use of both standard distance circle and minimum convex polygon is to ensure the robustness of the results, as a few studies have indicated that the circle and elliptic measures such as standard distance circle and standard deviational ellipse may overestimate activity space (Li and Tong, 2016). Note that all the distance is calculated using straight-line distance.

## 4. Methodology

### 4.1. Mobile phone data processing

Fig. 2 presents the mobile phone data processing workflow. The workflow consists of individual residential location identification, individual filtering and labeling, trip and activity identification and cleaning, and trip behavior indicators calculation. The next paragraphs go over the key points of this workflow.

First, we estimated the residential locations of mobile phone users by identifying the most connected signal towers at nighttime (10:00p.m. to 5:00 a.m.) over a one-month period. Since we had four non-consecutive months of mobile phone data, we computed four residential locations for each mobile phone user, one location per month. We only kept users who had the same locations within the four months and those locations were within a 3 km straight-line distance buffer of the new metro stations. Note that the positioning accuracy of cell phone signal records is around 250 m depending on the density of signal towers; what we mean by the same location is not that the latitudes and longitudes of the four inferred residential locations are exactly the same, but that the straight-line distance

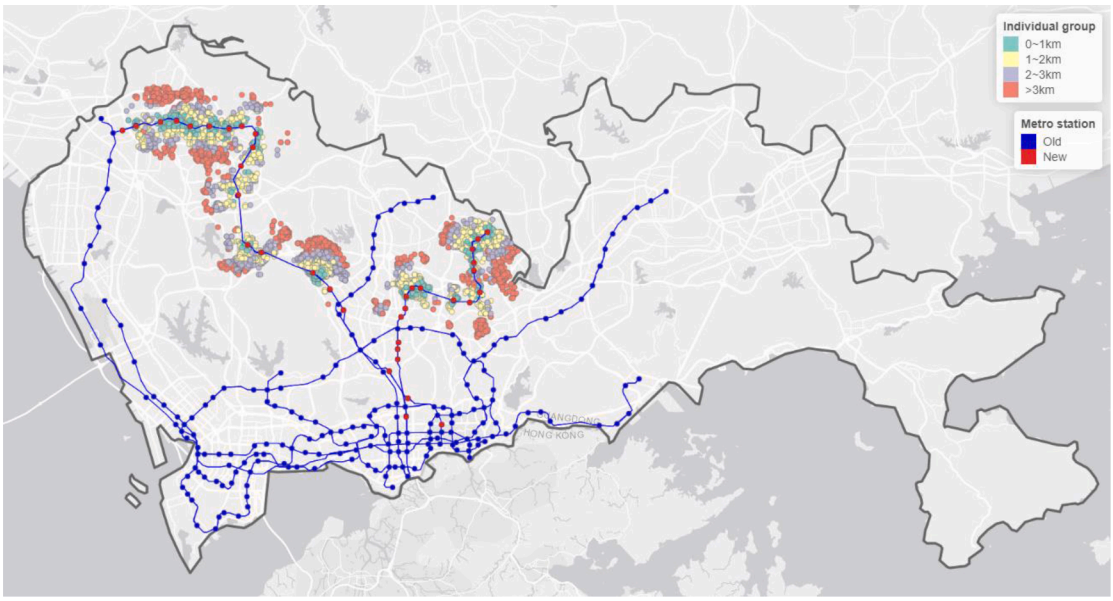


**Fig. 1.** The map of metro stations and residential locations of 7,799 individuals.

**Table 1**
Travel behavior indicators.

| Indicator | Description | Mean |
|---|---|---|
| Trip frequency | Total trip frequency for one month | 60.3 |
| Metro trip frequency | Total metro trip frequency for one month | 4.9 |
| Trip distance | Total trip distances for one month in km | 667.0 |
| Metro trip distance | Total metro trip distances for one month in km | 76.6 |
| Trip duration | Total trip duration for one month in min | 1942.0 |
| Metro Trip duration | Total metro trip duration for one month in min | 249.6 |
| Number of distinct visited locations | A count of distinct visited locations in one month | 15.7 |
| Standard distance circle | A circular area with a radius of the standard distance of the distinct visited locations in one month in $km^2$ | 382.5 |
| Minimum convex polygon | The smallest possible polygon that includes the distinct visited locations in one month in $km^2$ | 363.3 |

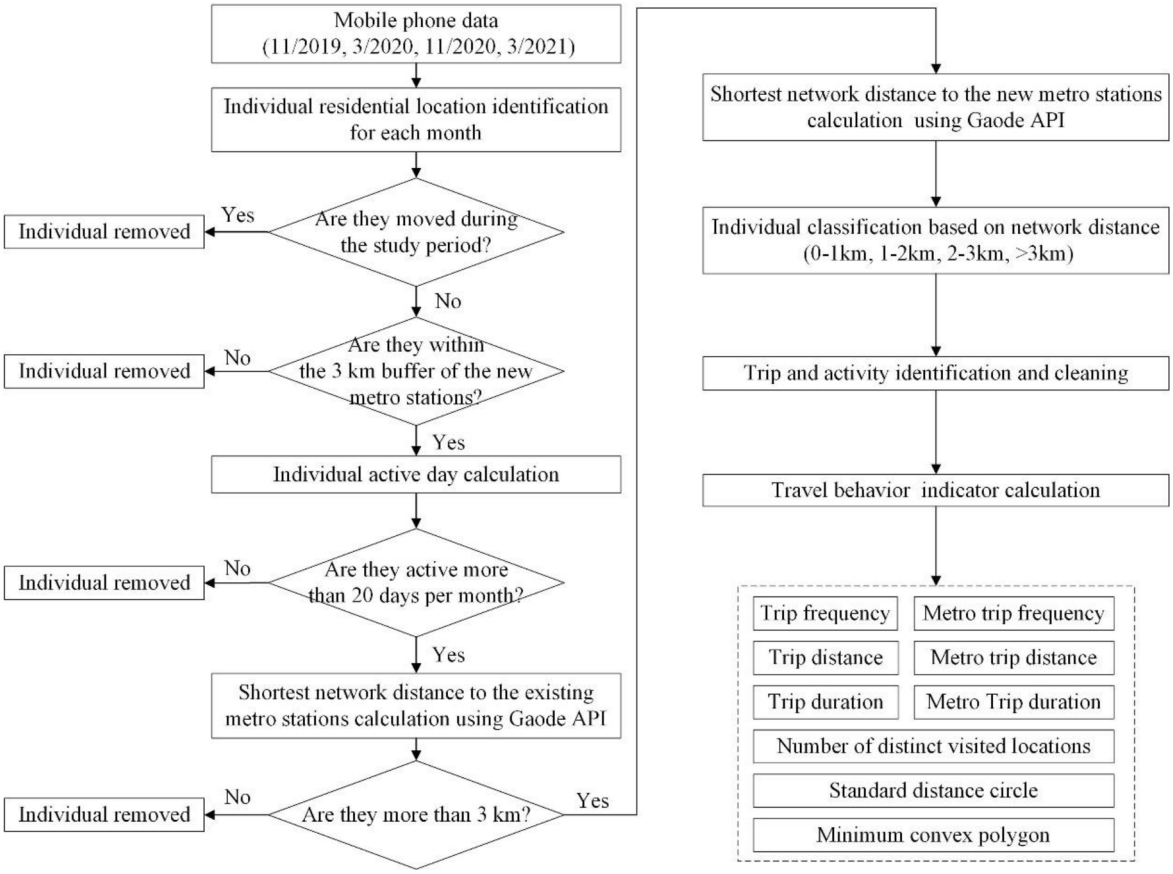


**Fig. 2.** Mobile phone data processing workflow.

between any two of the four locations is less than 250 m. Second, we counted the number of active days (the days with at least one trip record) for each mobile phone user. Only users who were active for more than 20 days in any one of the four months were retained. The retained users had sufficient records to ensure that the calculation of their travel behavior is reliable. Third, as the metro stations in Shenzhen's main urban area are very dense, some mobile phone users may already be within the catchment area of existing metro stations, which will affect the estimation of the effect of new metro stations. We acquired the network distance from their residential locations to the nearest existing metro stations from the Application Program Interface (API) of Gaode Map (the Chinese equivalent of Google Maps). We removed the users whose network distances from residential locations to existing metro stations were less than 3 km. Forth, we acquired the network distance from their residential locations to the nearest new metro stations using the same procedure as in step 2. The users were divided into four categories: 0–1 km, 1–2 km, 2–3 km, and > 3 km. Fifth, a small number of erroneous records of trips and activities existed in the mobile phone data. We developed three outlier removal criteria to clean the erroneous records: (1) trips with a duration less than 5 min or more than 180 min, (2) trips with a distance greater than 80 km, (3) trips with a speed less than 2 km/h or greater than 80 km/h. Sixth, we calculated the indicators of travel behavior by the month.

### 4.2. Panel data modeling

The data modeling is comprised of three stages. In the first stage, we used descriptive statistics and t-tests to analyze the differences in travel behavior indicators of treatment and control groups in the four months.

In the second stage, we built difference-in-differences regression for all travel behavior indicators. Difference-in-differences regression is widely used with panel data to estimate the effect of an intervention (the new metros in this case) by comparing the average differences in outcome variables between a treated group and a control group (Ellison et al., 2017). With $Y_{it}$ defined as the mean of the outcome for group $i$ at time $t$, the difference-in-differences estimator is $(Y_{11} - Y_{01}) - (Y_{10} - Y_{00})$, where $i = 1$ for the treatment group and $i = 0$ for the control group. The difference-in-differences estimator can be estimated with the following regression model:

$$Y_{it} = \beta_0 + \beta_1 X_i + \beta_2 T_t + \beta_3 X_i{*}T_t + \varepsilon_{it} \tag{3}$$

where $Y_{it}$ is the outcome for individual $i$ at time $t$, $X_i$ is a dummy variable with a value of 1 for the treatment group and 0 for the control group, and $T_t$ is a dummy variable with a value of 1 for the after opening period and 0 for the before opening period. Because $X_i{*}T_t$ only takes a value of 1 for treatment individuals in the after opening period, the coefficient $\beta_3$ on the interaction term represents the difference-in-differences estimator, i.e., the effect of the treatment on the treated individuals.

In the third stage, we performed a parallel trends test to ensure that the difference between the treatment and control groups remained constant over time before the intervention, i.e., the internal validity of difference-in-differences regression. We have two months of mobile phone data before the opening of the new metros (wave 1 and wave 2). We defined wave 2 as "After" and wave 1 as "Before". The parallel trends assumption is valid if the difference-in-differences estimator is insignificant. The graphical representation of the difference-in-differences regression and parallel trends test is shown in Fig. 3.

In this study, we have three treatment groups and three treatment time periods (two post-opening time periods for intervention effect estimation and one pre-opening time period for parallel trends testing). If using the base difference-in-differences regression directly, we would have to build nine sub-sample difference-in-differences regression models for each outcome variable. To reduce the number of difference-in-differences regression models, we built a full difference-in-differences regression model that can estimate nine interaction terms simultaneous in one model. For each outcome variable, we only needed one difference-in-differences regression model and did not need to subset the data. The formula is:

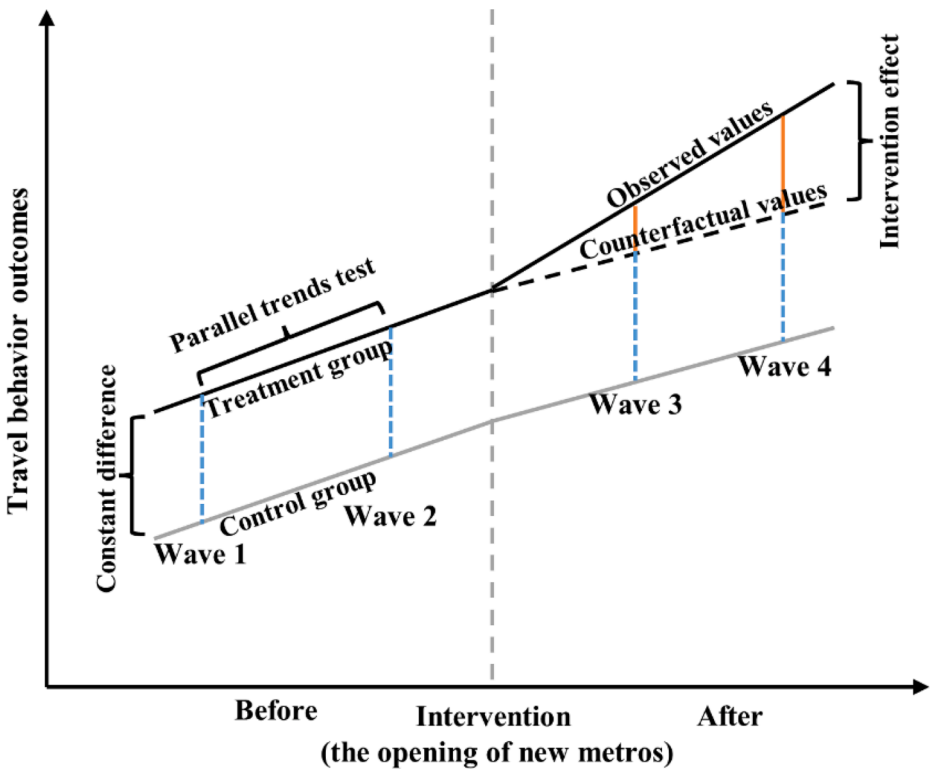


**Fig. 3.** The graphic representation of the difference-in-differences regression.

$$\begin{aligned} Y = \beta_0 + \beta_1 X_{w2} + \beta_2 X_{w3} + \beta_3 X_{w4} + \beta_4 X_{t1} + \beta_5 X_{t2} + \beta_6 X_{t3} + \\ \beta_7 X_{w2}{*}X_{t1} + \beta_8 X_{w2}{*}X_{t2} + \beta_9 X_{w2}{*}X_{t3} + \\ \beta_{10} X_{w3}{*}X_{t1} + \beta_{11} X_{w3}{*}X_{t2} + \beta_{12} X_{w3}{*}X_{t3} + \\ \beta_{13} X_{w4}{*}X_{t1} + \beta_{14} X_{w4}{*}X_{t2} + \beta_{15} X_{w4}{*}X_{t3} + \varepsilon \end{aligned} \tag{4}$$

where $Y$ is the travel behavior outcomes; $X_{t1}$, $X_{t2}$, and $X_{t3}$ are treatment/control dummy variables, coded (1, 0, 0) for the 0–1 km treatment group, (0, 1, 0) for the 1–2 km treatment group, (0, 0, 1) for the 2–3 km treatment group, and (0, 0, 0) for the control group; $X_{w2}$, $X_{w3}$, and $X_{w4}$ are time period dummy variables, coded (0, 0, 0) for wave 1 (nine months before opening), (1, 0, 0) for wave 2 (five months before opening), (0, 1, 0) for wave 3 (three months after opening), and (0, 0, 1) for wave 4 (seven months after opening). The nine coefficient estimates of the interaction terms ($\beta_7$, $\beta_8$, $\beta_9$, $\beta_{10}$, $\beta_{11}$, $\beta_{12}$, $\beta_{13}$, $\beta_{14}$, and $\beta_{15}$) are of interest. The first three coefficients indicate the validity of the parallel trends assumption for three treatment groups; the last six coefficients represent the treatment effect of the new metros for three treatment groups at two post-opening time periods.

## 5. Results

The original data contained approximately 876,000 mobile phone users (varied by months) whose residential locations were within a 3 km straight-line distance buffer of the new metro stations. During the data processing, 14.1% of users had the same residential locations within the four non-consecutive months. 10.0% of uses were active for more than 20 days in any one of the four months.

**Table 2**
Descriptive statistics of travel behavior indicators.

| Indicator | Wave | 0–1 km | | 1–2 km | | 2–3 km | | >3 km |
|---|---|---|---|---|---|---|---|---|
| | | Mean | t sig. | Mean | t sig. | Mean | t sig. | Mean |
| Trip frequency | 1 | 59.18 | 0.000*** | 60.28 | 0.002** | 60.19 | 0.001** | 62.40 |
| | 2 | 58.32 | 0.000*** | 59.20 | 0.002** | 59.63 | 0.012* | 61.36 |
| | 3 | 60.15 | 0.000*** | 60.60 | 0.000*** | 60.44 | 0.000*** | 63.14 |
| | 4 | 59.36 | 0.002** | 59.35 | 0.002** | 59.85 | 0.019* | 61.40 |
| Metro trip frequency | 1 | 4.39 | 0.002** | 3.79 | 0.203 | 3.62 | 0.448 | 3.38 |
| | 2 | 3.58 | 0.001** | 3.01 | 0.177 | 2.74 | 0.626 | 2.59 |
| | 3 | 8.55 | 0.000*** | 6.23 | 0.000*** | 5.11 | 0.018* | 4.11 |
| | 4 | 8.08 | 0.000*** | 5.75 | 0.000*** | 4.98 | 0.002** | 3.73 |
| Trip distance (km) | 1 | 653.03 | 0.038* | 646.86 | 0.011* | 644.17 | 0.006** | 678.61 |
| | 2 | 618.46 | 0.026* | 614.38 | 0.012* | 611.51 | 0.006** | 647.47 |
| | 3 | 653.51 | 0.081 | 643.08 | 0.011* | 639.18 | 0.005** | 675.65 |
| | 4 | 721.41 | 0.605 | 704.31 | 0.079 | 704.68 | 0.085 | 728.45 |
| Metro trip distance (km) | 1 | 70.12 | 0.002** | 59.75 | 0.256 | 56.24 | 0.636 | 53.73 |
| | 2 | 57.15 | 0.003** | 48.56 | 0.192 | 43.12 | 0.799 | 41.79 |
| | 3 | 125.63 | 0.000*** | 96.30 | 0.000*** | 81.07 | 0.027* | 65.88 |
| | 4 | 120.56 | 0.000*** | 89.27 | 0.000*** | 78.76 | 0.012* | 61.76 |
| Trip duration (min) | 1 | 1999.6 | 0.001** | 1988.8 | 0.000*** | 1987.1 | 0.000*** | 2114.1 |
| | 2 | 1830.7 | 0.001** | 1814.6 | 0.000*** | 1810.5 | 0.000*** | 1939.6 |
| | 3 | 1921.7 | 0.000*** | 1913.6 | 0.000*** | 1937.6 | 0.000*** | 2063.5 |
| | 4 | 1874.1 | 0.009** | 1847.5 | 0.001*** | 1876.9 | 0.013* | 1963.9 |
| Metro trip duration (min) | 1 | 256.99 | 0.002** | 218.71 | 0.258 | 205.55 | 0.658 | 197.04 |
| | 2 | 206.17 | 0.001** | 176.17 | 0.118 | 155.54 | 0.658 | 147.35 |
| | 3 | 371.75 | 0.000*** | 293.76 | 0.000*** | 269.96 | 0.022* | 221.56 |
| | 4 | 344.36 | 0.000*** | 259.39 | 0.001** | 246.68 | 0.010* | 196.54 |
| Number of distinct visited locations | 1 | 16.93 | 0.002** | 16.96 | 0.002** | 17.31 | 0.029* | 18.21 |
| | 2 | 12.75 | 0.001** | 12.87 | 0.003** | 13.13 | 0.025* | 13.99 |
| | 3 | 15.10 | 0.000*** | 15.35 | 0.003** | 15.51 | 0.009** | 16.58 |
| | 4 | 16.05 | 0.005** | 15.94 | 0.002** | 16.20 | 0.016* | 17.21 |
| Standard distance circle ($km^2$) | 1 | 397.54 | 0.629 | 385.71 | 0.477 | 386.00 | 0.496 | 392.79 |
| | 2 | 314.82 | 0.957 | 305.47 | 0.284 | 308.02 | 0.429 | 315.31 |
| | 3 | 369.26 | 0.663 | 374.10 | 0.347 | 371.59 | 0.516 | 364.70 |
| | 4 | 442.76 | 0.017* | 449.48 | 0.003** | 441.36 | 0.026* | 415.69 |
| Minimum convex polygon ($km^2$) | 1 | 383.63 | 0.349 | 376.26 | 0.11 | 382.37 | 0.299 | 393.79 |
| | 2 | 267.41 | 0.096 | 264.13 | 0.049* | 272.24 | 0.242 | 284.21 |
| | 3 | 336.33 | 0.273 | 333.88 | 0.192 | 344.78 | 0.75 | 348.31 |
| | 4 | 399.25 | 0.949 | 399.60 | 0.974 | 401.52 | 0.895 | 399.98 |
| Active days | 1 | 25.33 | 0.000*** | 25.52 | 0.156 | 25.42 | 0.011* | 25.65 |
| | 2 | 25.50 | 0.012* | 25.68 | 0.548 | 25.60 | 0.144 | 25.74 |
| | 3 | 25.65 | 0.030* | 25.72 | 0.174 | 25.53 | 0.000*** | 25.84 |
| | 4 | 25.85 | 0.433 | 25.80 | 0.173 | 25.84 | 0.389 | 25.93 |
| N | | 1801 (23%) | | 2640 (34%) | | 1682 (22%) | | 1676 (21%) |

Note: Wave 1, 2, 3, and 4 represent nine months pre-opening, five months pre-opening, three months post-opening, and seven months post-opening, respectively. 0–1 km, 1–2 km, and 2–3 km are treatment groups and > 3 km is the control group. ***, **, * denote paired significance of each treatment with the control group at the 0.001, 0.01, and 0.05 levels, respectively.

63.1% of users had a network distance of more than 3 km from their residential locations to the existing metro stations. We totally screened out 89.0% of the mobile phone users.

Table 2 shows the group means and between-group t-tests for travel behavior indicators in the treatment and control groups at four waves. The number of people in the 0–1 km, 1–2 km, 2–3 km, >3 km groups are well-balanced, accounting for 23%, 34%, 22%, and 21% of the total population, respectively. The average number of active days is 25.66. The difference in the number of active days between the treatment and control groups is small, ranging from 0.06 to 0.32. After the opening of the new metros (wave 3 and 4), differences in metro travel behaviors become much larger and more significant: individuals within 0–3 km network distance of the new metro stations significantly took more metro trips, longer metro distances, and longer metro trip duration. For the other indicators, no consistent pattern of pre-post changes was observed, which calls for further examination by difference-in-differences analysis.

Table 3 shows the results of difference-in-differences regressions. For each travel behavior indicator, wave and treat are control variables reflecting the systematic difference in travel behavior indicators between waves and between catchment areas, respectively. Significant differences between waves may be due to seasonal factors and significant differences between catchment areas differences may be due to differences in the built environment or socio-demographics. We mainly focus on six interaction terms (i.e., the 0–1 km, 1–2 km, and 2–3 km groups at Waves 3 and 4), which is the pure impact of the new metros on treatment groups. For trip frequency, distance, and duration, only two interaction terms (i.e., the 0–1 km group at Wave 4 for trip frequency and distance) are significantly positive, and the other sixteen interaction terms are insignificant. For metro trips' frequency, distance, and duration, all interaction terms are significantly positive. For the number of distinct visited locations, the six interaction terms are all insignificant. For the standard distance circle, the three interaction terms at wave 4 are significantly positive, while the three interaction terms at wave 3 are insignificant. For the minimum convex polygon, only the interaction term of the 1–2 km group at Wave 4 is significantly positive.

Table 3 also includes the results of the parallel trends test. The coefficients on the interaction terms of the 0–1 km, 1–2 km, and 2–3 km groups at wave 2 are all insignificant for all travel behavior indicators, indicating that the treatment and control groups have similar trends between waves 1 and 2. As a result, the parallel trends assumption holds.

**Table 3**
Results of difference-in-differences regression.

| | Trip frequency | Metro trip frequency | Trip distance | Metro trip distance | Trip duration | Metro trip duration | Number of distinct visited locations | Standard distance circle | Minimum convex polygon |
|---|---|---|---|---|---|---|---|---|---|
| Intercept | 62.4*** | 3.38*** | 678.61*** | 53.73*** | 2114.09*** | 197.04*** | 18.21*** | 392.79*** | 393.79*** |
| Treat 1 | −3.21*** | 1.01** | −25.58* | 16.39** | −114.44*** | 59.96** | −1.28** | 4.75 | −10.15 |
| Treat 2 | −2.12** | 0.41 | −31.75* | 6.02 | −125.31*** | 21.67 | −1.26** | −7.07 | −17.53 |
| Treat 3 | −2.21** | 0.25 | −34.44** | 2.51 | −127.02*** | 8.52 | −0.9* | −6.78 | −11.41 |
| Wave 2 | −1.04** | −0.79*** | −31.14*** | −11.93*** | −174.54*** | −49.68*** | −4.22*** | −77.47*** | −109.57*** |
| Wave 3 | 0.74 | 0.74*** | −2.96 | 12.15*** | −50.56** | 24.52** | −1.63*** | −28.09*** | −45.47*** |
| Wave 4 | −0.99* | 0.35* | 49.83*** | 8.03** | −150.14*** | −0.50 | −1.00*** | 22.9** | 6.19 |
| Wave 2 * Treat 1 | 0.18 | −0.02 | −3.43 | −1.03 | 5.55 | −1.14 | 0.05 | −5.25 | −6.65 |
| Wave 2 * Treat 2 | −0.04 | 0.01 | −1.34 | 0.75 | 0.36 | 7.14 | 0.13 | −2.77 | −2.56 |
| Wave 2 * Treat 3 | 0.48 | −0.09 | −1.52 | −1.19 | −2.00 | −0.33 | 0.04 | −0.51 | −0.56 |
| Wave 3 * Treat 1 | 0.23 | 3.42*** | 3.44 | 43.35*** | −27.39 | 90.23*** | −0.2 | −0.19 | −1.83 |
| Wave 3 * Treat 2 | −0.42 | 1.7*** | −0.82 | 24.40*** | −24.64 | 50.53*** | 0.03 | 16.47 | 3.09 |
| Wave 3 * Treat 3 | −0.49 | 0.75** | −2.03 | 12.68** | 1.08 | 39.88** | −0.17 | 13.67 | 7.88 |
| Wave 4 * Treat 1 | 1.17* | 3.34*** | 18.54* | 42.42*** | 24.63 | 87.86*** | 0.12 | 22.32* | 9.42 |
| Wave 4 * Treat 2 | 0.06 | 1.61*** | 7.61 | 21.50*** | 8.82 | 41.18** | −0.02 | 40.87*** | 17.15* |
| Wave 4 * Treat 3 | 0.65 | 1.01*** | 10.67 | 14.49*** | 39.94 | 41.63** | −0.11 | 32.45** | 12.96 |

Note: Treat 1, 2, and 3 represent the 0–1 km, 1–2 km, and 2–3 km groups, respectively. Wave 1, 2, 3, and 4 represent nine months pre-opening, five months pre-opening, three months post-opening, and seven months post-opening, respectively. ***, **, and * denote significance at the 0.001, 0.01, and 0.05 levels.

## 6. Discussion

People living in neighborhoods within 0–3 km of the new metro stations increased their metro trips' frequency, duration, and distance three and seven months after the opening of the new metros. Several previous studies, including those conducted by Zhu and Diao (2016) in Singapore and Spears et al. (2017) in Los Angeles, have confirmed that urban rail transit trips will increase after the opening of a new urban rail transit line. However, the impact of urban rail transit stations was limited to a 1 km buffer zone in their studies. We found that metro stations continue to have a significant impact in the 1–3 km range, although the impact on metro trips' frequency, duration, and distance in the 1–3 km range decreased than that in the 0–1 km range. We calculated the metro trip share for different catchment areas as shown in Table 4. After the opening of the new metros, the average increase of metro trip share was 7.1% for 0–1 km, 4.3% for 1–2 km, 3.1% for 2–3 km, and 1.5% for beyond 3 km. We further subdivide the areas>3 km into 3–4 km, 4–5 km, 5–8 km and found that the metro trip share is nearly stable. This confirms the rationality of choosing > 3 km as the control group.

We cannot differentiate the feeder modes using mobile phone data. However, the 2–3 km range is obviously not the distance of walking access (van Soest et al., 2019). Several questionnaire-based studies have revealed that bikes and buses are the primary modes of solving the first- and last-mile problems of metro travel in a longer range, while car transfer trips are extremely limited in China. The popularity of bike share in China has further facilitated the use of bikes as a feeder mode (Deng and Zhao, 2021). Guo and He (2021) found that in Shenzhen, walking has the highest share of access and egress trips (66–72%), followed by bike share (12–14%). Bus is the third popular feeder mode (9–13%). Private bikes, taxi or ride-hailing, and other modes all have extremely small shares. Average Euclidean distances of feeder trips are approximately 0.8 km for bike share and 2.2 km for buses. If we scaled up the Euclidean distance to road-network distance by 1.47 (Shen et al., 2018), the average trip distance is 1.2 km for bike share and 3.2 km for buses. Zhao and Li (2017) found that the mean distance for bikes to metro stations was about half of the mean distances for buses in Beijing.

The total trip frequency and distance of people in the 0–1 km range increased significantly seven months after opening although it did not change significantly three months after opening. The increase in total trip frequency and distance is due to the increase in metro trips. However, although the new metros have an impact on metro trips in the 0–3 km range, the new metros only have a significant impact on total trip frequency and distance in the 0–1 km range, implying that the hypothesis of induced travel demand only conditionally holds. According to Ibraeva et al. (2021), it takes some time for people to adjust their total trips to the new transportation option. Total trip duration did not change significantly after the introduction of the new metros. This is consistent with the concept of the constant travel time budget, i.e., that a large group of people at the aggregate level spends a quite constant proportion of their total time traveling, 60–75 min per person per day (Marchetti's constant) (Mokhtarian and Chen, 2004). Although trip frequency and distance increased in some treatment groups, the modal shifts of some trips from the bus and nonmotorized modes to the metro increased average delivery speed, so the total trip duration did not change significantly.

Identifying the source of ridership for new metro projects has important policy implications. Possible sources include shifts from other transportation modes and induced travel demand as a result of the new metro. Since the new metros have only induced new travel demand in the 0–1 km range seven months after opening, the main source of metro ridership is transportation mode shifts. We assumed that every mobile phone user adopts a single transportation mode in his/her trips and used simple rules to differentiate walking, bike, e-bike, bus, and car trips from total trips. We assumed that 6 km/h is the maximum speed of walking, 6–15 km/h is the speed of bikes, 15–30 km/h is the speed of buses and e-bikes, and 30 km/h is the minimum speed of cars. Most of these speed thresholds have been adopted in previous studies (Huang et al., 2019; Qu et al., 2015). E-bike and bus modes were not differentiated as their speeds are largely overlapped (Lin et al., 2008). We scaled up the Euclidean distance to road-network distance by 1.47 (Shen et al., 2018). Although these simple rules cannot achieve very accurate results for single trips, our tests confirmed that the aggregated mode share derived using these simple rules can closely match the results of the official household travel survey in Shenzhen. The changes and rates of changes in trips frequency, distance, and duration for all transportation modes are shown in Table 5. For example, the change in metro trips frequency is calculated as the average metro trip frequency in Wave 3 and 4 minus that in Wave 1 and 2. The rate of change in metro trip frequency is calculated as the average metro trip share in Wave 3 and 4 minus that in Wave 1 and 2.

In general, the trip frequency, distance, and duration for metro and car modes increased, while those for walking, bike, e-bike & bus decreased. Several studies, including Lee and Senior (2013) in the United Kingdom and Wu and Hong (2017) in China, have suggested that the increase in urban rail transit ridership is primarily due to bus and nonmotorized trips, rather than cars trips. However, Dai et al. (2020) and Zhu and Diao (2016) argued that the private car mode share decreased after the opening of a new metro line in Singapore.

**Table 4**
Metro trip shares.

| | 0–1 km | 1–2 km | 2–3 km | >3 km | 3–4 km | 4–5 km | 5–8 km |
|---|---|---|---|---|---|---|---|
| Wave 1 | 7.4% | 6.3% | 6.0% | 5.4% | 5.2% | 6.6% | 3.8% |
| Wave 2 | 6.1% | 5.1% | 4.6% | 4.2% | 4.2% | 4.9% | 2.6% |
| Wave 3 | 14.2% | 10.3% | 8.5% | 6.5% | 6.4% | 7.3% | 5.4% |
| Wave 4 | 13.6% | 9.7% | 8.3% | 6.1% | 5.8% | 7.3% | 4.5% |
| Average change | 7.1% | 4.3% | 3.1% | 1.5% | 1.4% | 1.5% | 1.7% |

Note: Wave 1, 2, 3, and 4 represent nine months pre-opening, five months pre-opening, three months post-opening, and seven months post-opening, respectively. 0–1 km, 1–2 km, and 2–3 km are treatment groups and > 3 km is the control group. 0–1 km, 1–2 km, and 2–3 km are treatment groups and > 3 km is the control group. The average change is calculated as $\frac{Wave4 + Wave3}{2} - \frac{Wave2 + Wave1}{2}$.

**Table 5**
Transportation mode changes.

| | Mode | 0–1 km | 1–2 km | 2–3 km | >3 km |
|---|---|---|---|---|---|
| Trip frequency | Metro | 4.33 (7.1%) | 2.59 (4.3%) | 1.87 (3.1%) | 0.94 (1.5%) |
| | Walking | −1.11 (−2.0%) | −0.89 (−1.5%) | −0.93 (−1.6%) | −0.96 (−1.6%) |
| | Bike | −2.89 (−5.4%) | −2.41 (−4.2%) | −1.94 (−3.4%) | −1.35 (−2.4%) |
| | Bus & e-bike | −0.77 (−1.9%) | −0.40 (−0.8%) | −0.05 (−0.2%) | 0.11 (0.0%) |
| | Car | 1.45 (2.2%) | 1.34 (2.2%) | 1.29 (2.1%) | 1.65 (2.6%) |
| Trip distance (km) | Metro | 59.46 (8.0%) | 38.63 (5.3%) | 30.23 (4.0%) | 16.06 (1.9%) |
| | Walking | −3.61 (−0.7%) | −2.81 (−0.6%) | −3.00 (−0.6%) | −2.88 (−0.6%) |
| | Bike | −22.69 (−5.2%) | −17.23 (−4.1%) | −14.15 (−3.8%) | −10.72 (−3.0%) |
| | Bus & e-bike | −5.10 (−3.7%) | 1.68 (−2.2%) | 9.17 (−1.1%) | 9.31 (−0.8%) |
| | Car | 23.67 (1.6%) | 22.82 (1.7%) | 21.83 (1.5%) | 27.24 (2.4%) |
| Trip duration (min) | Metro | 126.47 (6.8%) | 79.14 (4.3%) | 77.77 (4.1%) | 36.85 (1.9%) |
| | Walking | −49.12 (−2.5%) | −37.64 (−1.8%) | −40.05 (−2.1%) | −40.41 (−1.9%) |
| | Bike | −106.00 (−5.2%) | −85.60 (−4.1%) | −68.80 (−3.8%) | −54.94 (−2.5%) |
| | Bus & e-bike | −14.42 (−0.6%) | −1.53 (0.2%) | 15.32 (0.6%) | 14.76 (0.9%) |
| | Car | 25.82 (1.5%) | 24.44 (1.4%) | 24.18 (1.3%) | 30.66 (1.6%) |

Note: 0–1 km, 1–2 km, and 2–3 km are treatment groups and > 3 km is the control group.

We found that the increase in metro ridership was primarily due to bus and nonmotorized trips. Although metro trips did not replace car trips, the metro is still important in the context of motorization (walking and bike trips decrease while bus, e-bike, metro, and car trips increase in the control group), because bus and nonmotorized trips are more likely to be replaced by driving if metro is not available as an alternative (car trips increase less in the treatment groups).

The new metro stations had an insignificant effect on the standard distance circle and minimum convex polygon three months after opening, but had a significant positive effect on the standard distance circle and minimum convex polygon seven months after opening, which shows activity behavior adaptations require more time than travel behavior adaptations. Time geography notes that people select activity and travel options within the boundaries of capability constraints, coupling constraints, and authority constraints (Hägerstrand, 1970). The new metro reduces capability constraints and increases time–space flexibility therefore people's activity space expands. Botte and Olaru (2012) suggested that the standard distance circle or standard deviational ellipse may exaggerate the size of the activity space, especially for people with fewer reported visited locations. We also found that the increase in the standard distance circle is larger and more significant than the increase in the minimum convex polygon. Surprisingly, people in the 1–2 km and 2–3 km range increased their activity space (either measured by the standard distance circle or minimum convex polygon) more than those in the 0–1 km range. The overall structure of the activity spaces can be divided into three parts: home-based sub-activity space, workplace-based sub-activity space, and commuting-based sub-activity space (Li and Tong, 2016). The new metro may affect all three sub-activity spaces for metro riders, but what differentiates people in the 0–1 km, 1–2 km, and 2–3 km range is most likely home-based sub-activity space. The home-based secondary activity space of people in the 1–2 km and 2–3 km range may initially be around their residential locations and will expand to include the area around the metro stations, whereas the home-based sub-activity space of people in the 0–1 km range may not change significantly because the new metro stations are within their initially home-based sub-activity space.

## 7. Conclusion

This is the first study to use large-scale panel data to assess the impact of the new metro on travel behavior and provide insights into causality. The key findings are that the catchment area of the metro stations could be expanded to 3 km for metro trips and activity space. The new metro had a significant impact on total trip frequency and distance only in the 0–1 km range. The increase in metro ridership was primarily due to modal shifts from the bus and nonmotorized trips. Furthermore, this study confirms that mobile phone data have great potential for panel analysis of travel behavior.

There are also some practical implications. First, when studying the impact of the metro or conducting TOD planning, a larger impact range should be used to maximize the utility of the metro. High-density TOD development in China's already densely populated cities results in the less comfortable living environment. Cities and residents would benefit more by promoting TOD in larger catchment areas rather than focusing solely on denser populations and jobs in traditional 0–1 km catchment area. Second, the expansion of the impact range relies on the use of bikes, especially bike share, and buses. Bikes and buses can be better leveraged as metro feeder modes by developing dedicated bike lanes and bike parking, planning circular bus routes, and coordinating bus and metro schedules. Third, although metro trips did not replace car trips, the metro is still important because bus and nonmotorized trips are more likely to be replaced by driving if the metro is not available as an alternative in the context of motorization. Fourth, in this study, the positive changes in travel behavior are due to the new metro rather than the TOD around the new metro stations, because changes in TOD-related features (density or mixed uses) require more time. Transforming adjacent neighborhoods into transit-oriented, more compact, mixed-use communities will expand the metro's positive effect even further.

*CRediT authorship contribution statement*

**Yiling Deng:** Conceptualization, Data curation, Formal analysis, Funding acquisition, Methodology, Writing – original draft. **Pengjun Zhao:** Conceptualization, Funding acquisition, Writing – review & editing.

## Declaration of Competing Interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

## Data availability

Data will be made available on request.

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